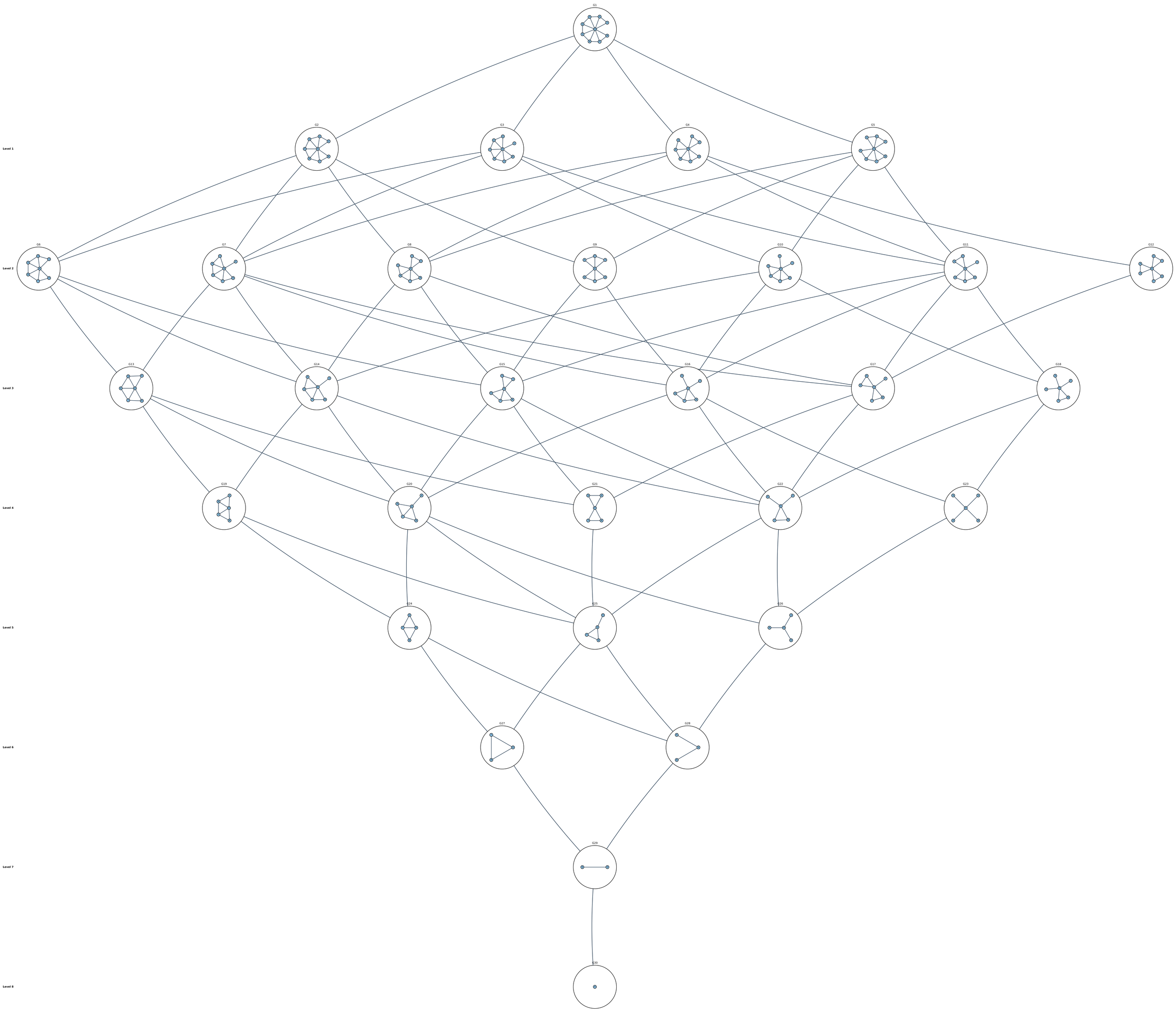Edge Contraction Spectrum $\Sigma(F_8)$